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# EduOS Cameroon — Monitoring & Evaluation Framework and Results Matrix

|             |                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Document ID | EDUOS-MEF-001                                                                                                                             |
| Version     | 1.0                                                                                                                                       |
| Structure   | Theory of change → results matrix (indicator, baseline, source, targets by year) → measurement & verification arrangements                |
| Baselines   | From Baseline Data Annex (EDUOS-BDA-001); baselines marked TBD-P are established during the Phase I pilot and frozen at the Phase II gate |

## 1. Theory of change

**IF** every approved textbook title and copy has a digital identity tracked from print to learner (activities), **THEN** ministries see real stock, real coverage, and real losses (outputs), **THEN** procurement matches need, distribution reaches classrooms, and books live their full service life (outcomes), **THEREFORE** more learners hold textbooks for more of the school year at lower cost per learner-book-year (impact) — protecting the ERSP investment that moved Cameroon from 12:1 to 3-books-per-2-learners.

Critical assumptions (monitored as risk indicators, §4): dual-ministry governance holds (R1); schools adopt the tools (R3); recurrent financing lands in the MTEF (R5).

## 2. Results matrix

Years are programme years (Y1 = first build year). Targets are cumulative and verified by the source shown; “system” = platform analytics with independent annual data audit (Budget line 8).

**Disaggregation rule (EDUOS-ESS-001 INC-01):** every coverage and access indicator (IMP-1/2, OUT-1, OUT-P2/P5) is reported disaggregated by region, urban/rural, school ownership, and single-sex/mixed status — and by learner sex where student-level assignment is active. OUT-P5 additionally tracks the female share of trained staff against the female share of school leadership (INC-03 floor).

**IMPACT LEVEL**

| <b>ID</b> | <b>Indicator</b>                                                               | <b>Baseline</b>                                        | <b>Y3</b>  | <b>Y5</b> | <b>Y7</b> | <b>Source / verification</b>                         |
|-----------|--------------------------------------------------------------------------------|--------------------------------------------------------|------------|-----------|-----------|------------------------------------------------------|
| IMP-1     | Pupil-textbook ratio, essential subjects, public primary (national)            | 1.5 books/learner (2023, WB)                           | ≥1.5 held  | ≥2.0      | ≥2.5      | EMIS/system + annual school sample survey            |
| IMP-2     | Pupil-textbook ratio in bottom-decile schools (equity)                         | up to 30:1 pre-reform pockets; pilot-measured<br>TBD-P | TBD-P +25% | ≥1:1      | ≥1.5:1    | system coverage report RPT-COV + verification survey |
| IMP-3     | Cost per learner-book-year (procurement flow ÷ effective book-years delivered) | derived ≈ 1,230 FCFA (ECO §2–3); pilot-verified        | –5%        | –15%      | –20%      | procurement records + system service-life data       |

## OUTCOME LEVEL

| ID    | Indicator                                                                                   | Baseline                                          | Y2<br>(pilot)         | Y3                              | Y5            | Source                                   |
|-------|---------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------|---------------------------------|---------------|------------------------------------------|
| OUT-1 | % of procured textbooks confirmed received at destination school (anti-leakage, benefit B1) | not measurable today (0% traced)                  | ≥90% in pilot regions | ≥92%                            | ≥97% national | system passport events, audited          |
| OUT-2 | Median days from warehouse dispatch to school receipt confirmation                          | TBD-P (pilot baseline)                            | measure               | –25%                            | –50%          | system timestamps                        |
| OUT-3 | Forecast accuracy:                                                                          | forecast – actual enrolment                       | aggregated nationally | TBD-P vs current cascade method | measure       | ≤8% error                                |
| OUT-4 | Average textbook service life (essential titles)                                            | 3 yrs design; actual TBD-P                        | measure               | +0.25 yr                        | +0.5 yr       | condition/retirement events (benefit B3) |
| OUT-5 | Annual verification campaign: % of expected copies accounted for                            | 0% (no campaigns exist)                           | ≥80% pilot            | ≥85%                            | ≥95%          | FR-NTR-12 campaign reports               |
| OUT-6 | School staff time on textbook administration                                                | TBD-P (20-school time-and-motion study, ECO §8.3) | baseline set          | –30%                            | –50%          | repeat study (benefit B4)                |

## OUTPUT LEVEL

| ID     | Indicator                                                              | Baseline | Y1                              | Y2                     | Y3                       | Y5                        | Source                                         |
|--------|------------------------------------------------------------------------|----------|---------------------------------|------------------------|--------------------------|---------------------------|------------------------------------------------|
| OUT-P1 | Approved titles registered with NTID                                   | 0        | 100% of active catalogue        | maintain               | maintain                 | maintain                  | RPT-CAT                                        |
| OUT-P2 | Public schools live on the platform (≥1 sync in last 90 days)          | 0        | 500 (pilot)                     | 3,000                  | 8,000                    | 18,500                    | sync telemetry                                 |
| OUT-P3 | % of textbook movements captured digitally in live schools             | 0%       | ≥95% (acceptance test, FRS §12) | ≥95%                   | ≥95%                     | ≥95%                      | passport event coverage vs procurement records |
| OUT-P4 | Copies/batches under passport tracking                                 | 0        | pilot print runs                | 100% of new print runs | + brown-field stock ≥50% | ≥90% of circulating stock | batch registration (FR-NTR-06, MIG-02)         |
| OUT-P5 | School staff trained (2 per school) and certified                      | 0        | 1,000                           | 6,000                  | 16,000                   | 37,000                    | training academy records                       |
| OUT-P6 | Off-grid live schools with solar kits installed                        | 0        | pilot subset                    | 1,500                  | 3,500                    | 5,500                     | asset register                                 |
| OUT-P7 | Data-quality score of School Registry seed (completeness × validation) | TBD (R8) | ≥85% pilot regions              | ≥90%                   | ≥90%                     | ≥95%                      | data-audit                                     |

**SYSTEM HEALTH (CONTINUOUS, FROM PLATFORM SLOS)**

| ID    | Indicator                                          | Target                               | Source               |
|-------|----------------------------------------------------|--------------------------------------|----------------------|
| SYS-1 | Central service availability                       | ≥99.5% monthly (NFR-NTR-04)          | monitoring           |
| SYS-2 | Devices syncing within 30 days                     | ≥90% of live schools                 | sync telemetry       |
| SYS-3 | Quarantined sync conflicts resolved within 30 days | ≥95% (FR-NTR-SYNC-01)                | reconciliation queue |
| SYS-4 | Passport hash-chain verification failures          | 0 unresolved > 7 days (FR-NTR-DM-02) | nightly audit job    |

**3. Measurement & verification arrangements**

1. **Self-reporting bias control.** Platform data is the primary source, but OUT-1, OUT-5, IMP-1/2 are verified annually by an **independent third party** sampling ≥400 schools (stratified by region, urban/rural, on/off-grid) — funded under Budget line 8. Divergence >5 pp between system and survey triggers a data-quality investigation, not target re-negotiation.
2. **Baseline freeze.** All TBD-P baselines are measured during the Y2 pilot and frozen in a Baseline Report approved at the Phase II gate; targets adjust only by Steering Committee decision recorded in the decision log.
3. **Phase-gate linkage.** Disbursement gates (BUD §6): Phase II requires OUT-P3 ≥95% and OUT-1 ≥90% in the pilot; Phase III requires OUT-P2 ≥8,000 and SYS-2 ≥90%. Failing a gate pauses expansion and triggers the corrective-support protocol (R3), not silent target erosion.
4. **Economic model re-estimation.** At each gate, the Economic Analysis (EDUOS-ECO-001) is re-run with measured values for B1–B4; results reported to financiers.
5. **Public transparency.** IMP and OUT indicators publish annually in an open national dashboard (catalogue API is already public, FR-NTR-13); school-identifiable performance data is used for support targeting, not published rankings, in Years 1–3 (R3).

## 4. Risk-assumption indicators (monitored quarterly)

| Assumption            | Indicator                                                                | Red line                                        |
|-----------------------|--------------------------------------------------------------------------|-------------------------------------------------|
| Governance holds (R1) | Steering Committee meets with quorum; shared-registry decisions ≤60 days | 2 consecutive missed quarters                   |
| Adoption (R3)         | SYS-2 trend; % trained schools active after 90 days                      | <70% activation                                 |
| Financing (R5)        | Recurrent line in MTEF by Y4 budget circular                             | absent at Y4 → Phase IV de-scope plan activates |
| Security context (R6) | Accessible-school ratio in NW/SW/Far North                               | rollout plan re-phased                          |

## 5. Evaluation calendar

| Event                                                    | Timing  | Owner                                       |
|----------------------------------------------------------|---------|---------------------------------------------|
| Baseline Report (pilot)                                  | end Y2  | PMU + independent verifier                  |
| Mid-term evaluation                                      | Y3 gate | external evaluator (financier-commissioned) |
| Time-and-motion repeat study                             | Y3, Y5  | independent verifier                        |
| Final evaluation & sustainability review                 | Y5      | external evaluator                          |
| Post-programme impact evaluation (optional, recommended) | Y7      | financier/research partner                  |